



FATE MONTHLY REPORT

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FORECAST SYSTEM FOR ATMOSPHERE AND TURBULENCE at the VERY LARGE TELESCOPE

Performance assessment
October 2023



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1 Title for Night 1

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1.1 Title for Figures of Night 1

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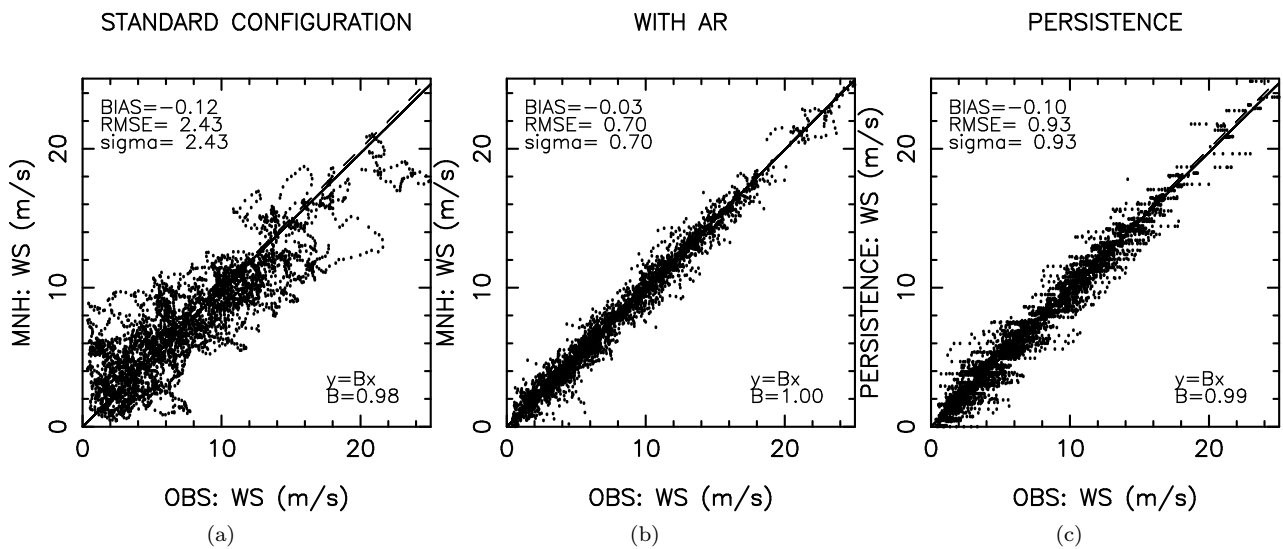


Figure 1: night1 - wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

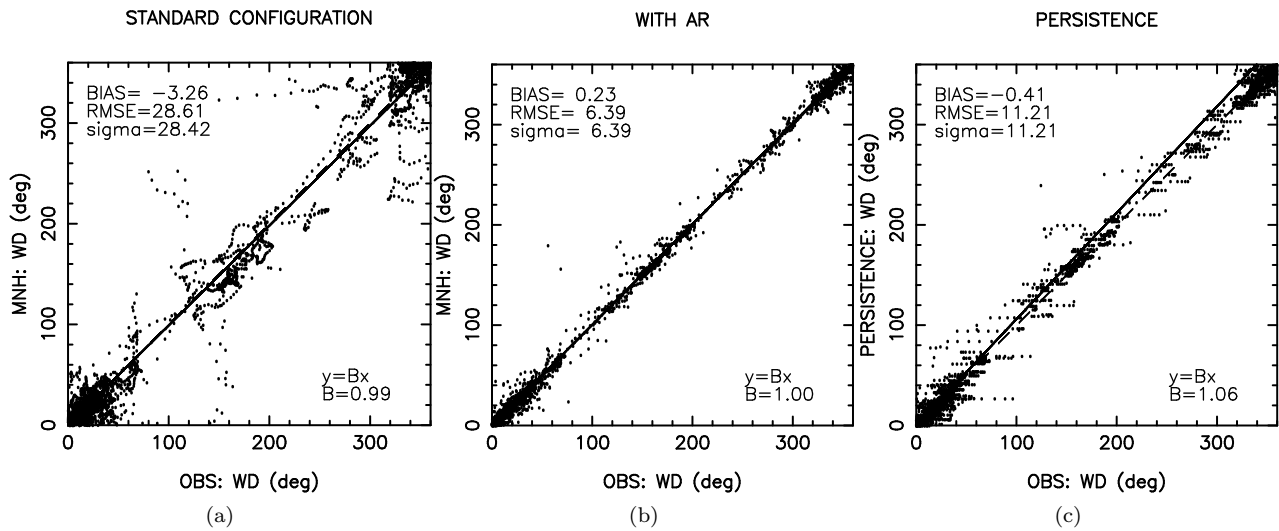


Figure 2: night1 - wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

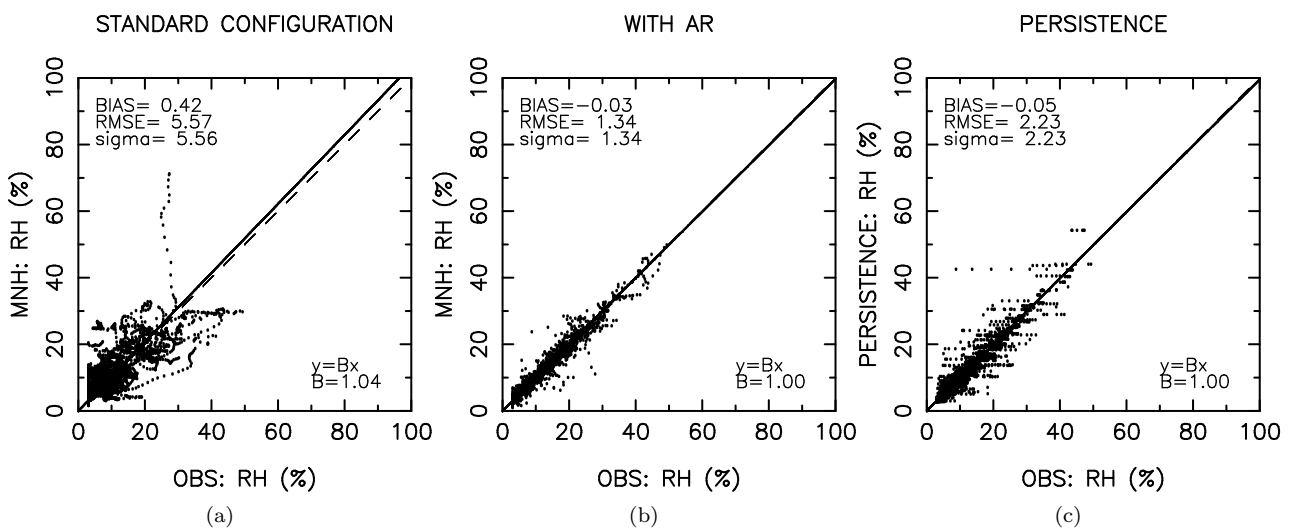


Figure 3: night1 - relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

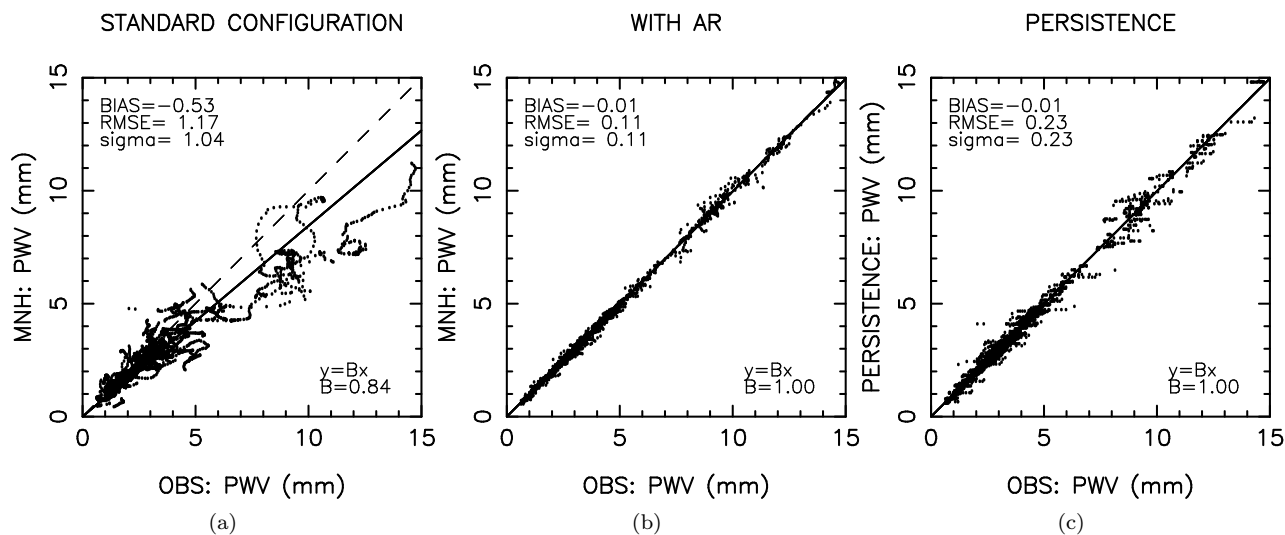


Figure 4: night1 - precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

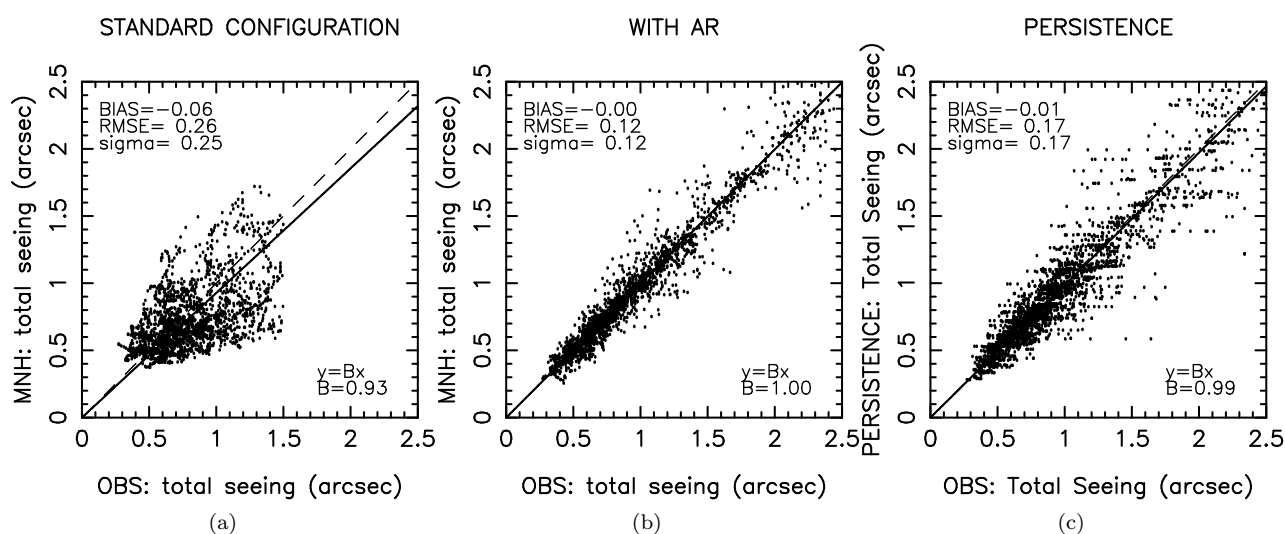


Figure 5: night1 - total seeing (arcsec): (a) STANDARD CONFIGURATION (< 1.5"), (b) WITH AR (1H), (c) PERSISTENCE (1H)

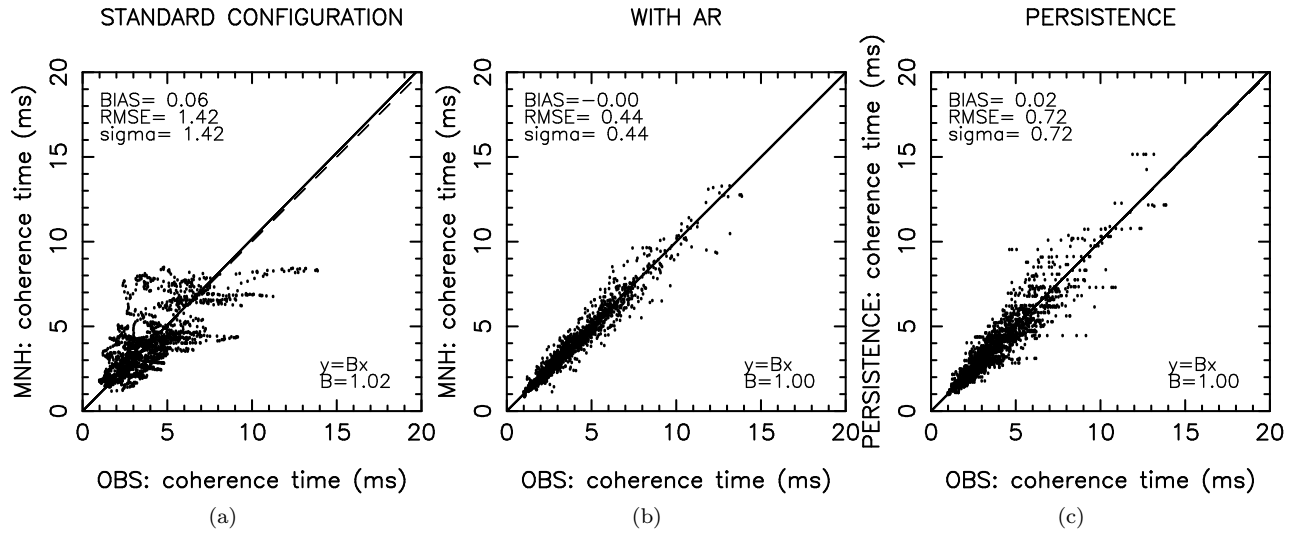


Figure 6: night1 - coherence time (ms): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

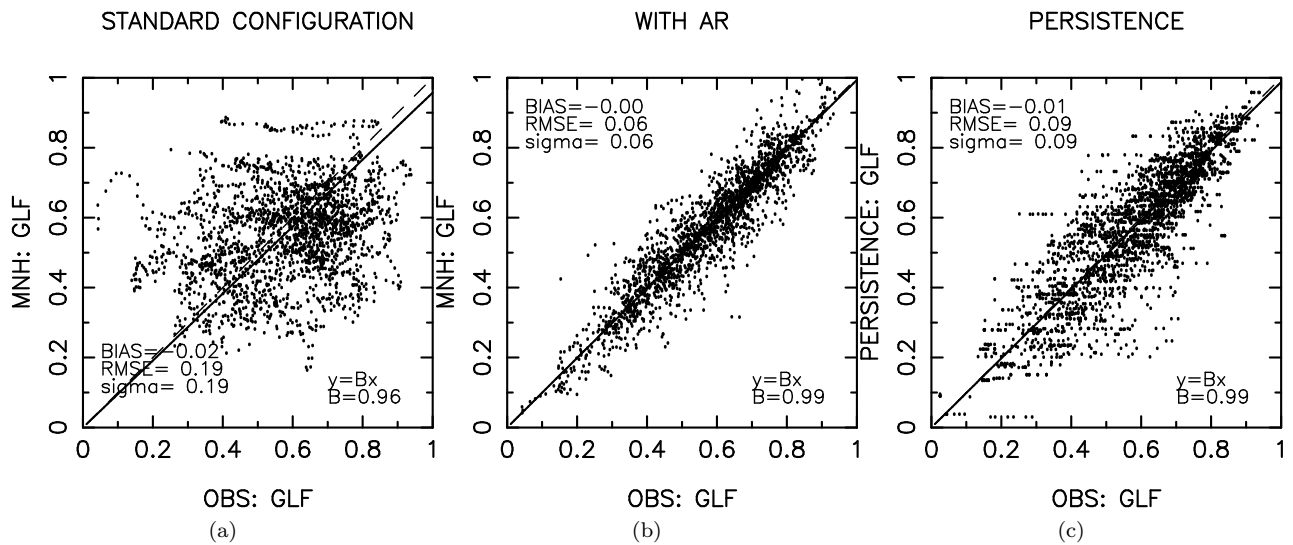


Figure 7: night1 - ground layer fraction (dimensionless): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

1.2 Title for Statistics of Night 1

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VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS (m·s ⁻¹)	-0.12	2.43	2.43	-0.76	2.25	2.12
WD (degree)	-3.26	28.61	28.42	-2.14	26.84	26.75
RH (percent)	0.42	5.57	5.56	0.03	6.13	6.13
PWV (mm)	-0.53	1.17	1.04	-0.75	1.51	1.32
SEE (arcsec)	-0.06	0.26	0.25	-0.01	0.24	0.24
TAU (ms)	0.06	1.42	1.42	-0.32	1.32	1.28
GLF (dimensionless)	-0.02	0.19	0.19	-0.09	0.19	0.16

Table 1: Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (October-2023)

FATE - Monthly Report

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.04	0.68	0.68		0.91	0.90
WD (degree)	0.11	6.59	6.58		11.52	11.46
RH (percent)	-0.03	1.54	1.54		2.75	2.75
PWV (mm)	-0.01	0.14	0.14		0.27	0.27
SEE (arcsec)	0.01	0.12	0.12		0.18	0.18
TAU (ms)	-0.01	0.48	0.48		0.81	0.81
GLF (dimensionless)	-0.01	0.06	0.06		0.09	0.09

Table 2: Last month (October-2023): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.03	0.70	0.70		0.93	0.93
WD (degree)	0.23	6.39	6.39		11.21	11.21
RH (percent)	-0.03	1.34	1.34		2.23	2.23
PWV (mm)	-0.01	0.11	0.11		0.23	0.23
SEE (arcsec)	-0.00	0.12	0.12		0.17	0.17
TAU (ms)	-0.00	0.44	0.44		0.72	0.72
GLF (dimensionless)	-0.00	0.06	0.06		0.09	0.09

Table 3: Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

PARAMETER	STANDARD	WITH AR (1H)
POD1	60.1	94.2
POD2	55.0	86.9
POD3	87.7	98.4
PC	71.2	94.2
EBD	1.3	0.0

Table 4: night1 - PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	98.7	98.7
POD2	48.2	90.1
POD3	46.7	88.6
PC	89.0	97.0
EBD	0.0	0.0

Table 5: night1 - PODs for wind speed (ms^{-1}) and thresholds (12.0,18.0) ms^{-1}

1.3 Title for PODs tables of Night 1

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PARAMETER	STANDARD	WITH AR (1H)
POD1	93.8	99.2
POD2	46.3	84.7
POD3	87.6	97.9
POD4	44.4	97.0
PC	86.5	97.7
EBD	1.1	0.0

Table 6: night1 - PODs for wind direction (degree)

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	49.7	92.6
POD2	69.0	94.3
POD3	80.6	93.9
PC	65.7	93.7
EBD	1.5	0.1

Table 7: night1 - PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.7	100.0
POD2	NaN	NaN
POD3	NaN	NaN
PC	99.7	100.0
EBD	0.1	0.0

Table 8: night1 - PODs for relative humidity (percent) and thresholds (50,70) percent

PARAMETER	STANDARD	WITH AR (1H)
POD1	95.8	98.3
POD2	78.9	97.3
POD3	72.7	98.2
PC	82.3	97.9
EBD	0.0	0.0

Table 9: night1 - PODs for precipitable water vapor (mm)

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	58.7	88.4
POD2	41.8	76.0
POD3	47.8	95.2
PC	48.4	87.9
EBD	5.7	0.0

Table 10: night1 - PODs for total seeing (arcsec) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	93.2	100.0
POD2	95.7	99.7
POD3	87.1	100.0
PC	94.0	99.9
EBD	4.6	0.0

Table 11: night1 - PODs for total seeing (arcsec) and accuray 0.24

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	73.3	96.7
POD2	59.4	86.8
POD3	65.0	90.4
PC	67.4	92.4
EBD	1.4	0.0

Table 12: night1 - PODs for coherence time (ms) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	94.0	100.0
POD2	88.7	99.9
POD3	64.2	98.7
PC	89.9	99.9
EBD	4.6	0.1

Table 13: night1 - PODs for coherence time (ms) and accuray 1.22

FATE - Monthly Report

PARAMETER	STANDARD	WITH AR (1H)
POD1	60.8	91.0
POD2	48.6	72.1
POD3	16.2	83.0
PC	46.2	82.6
EBD	12.9	0.1

Table 14: night1 - PODs for ground layer fraction (dimensionless) and accuray 0.0

PARAMETER	STANDARD	WITH AR (1H)
POD1	87.7	99.9
POD2	88.2	99.0
POD3	48.3	99.2
PC	81.8	99.5
EBD	13.7	0.1

Table 15: night1 - PODs for ground layer fraction (dimensionless) and accuray 0.14

2 Title for Day 1

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2.1 Title for Figures of Day 1

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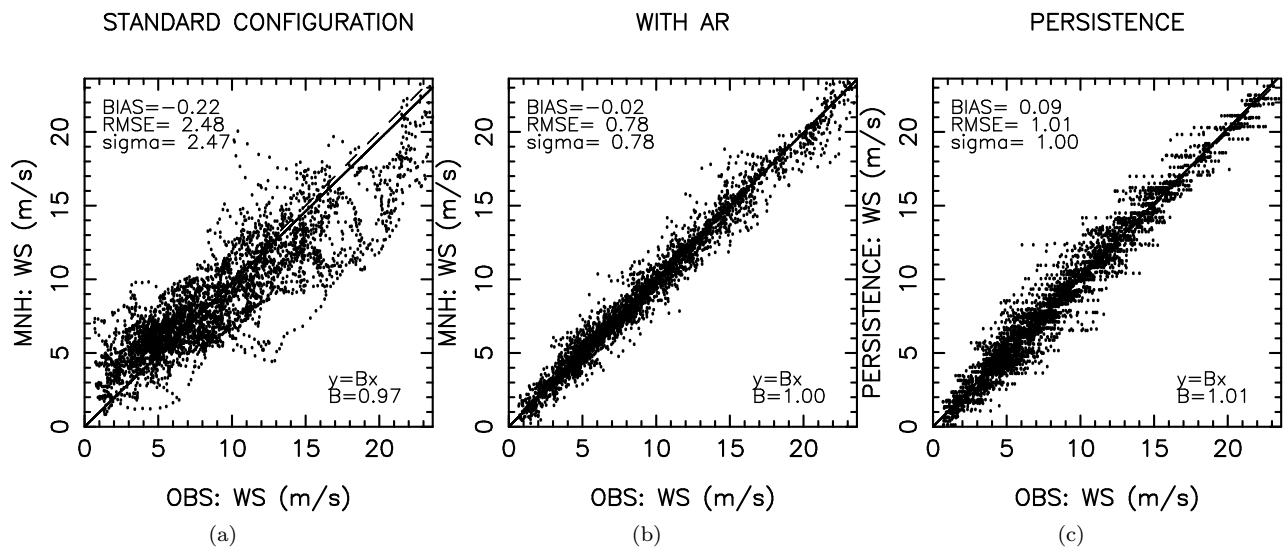


Figure 8: day1 - wind speed (ms^{-1}): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

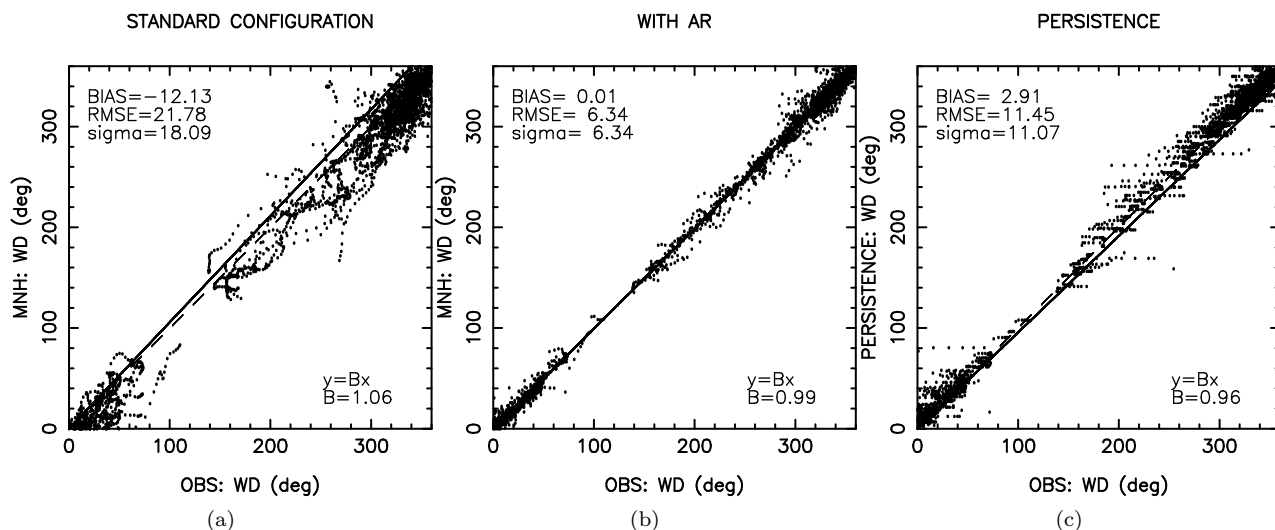


Figure 9: day1 - wind direction (degree): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

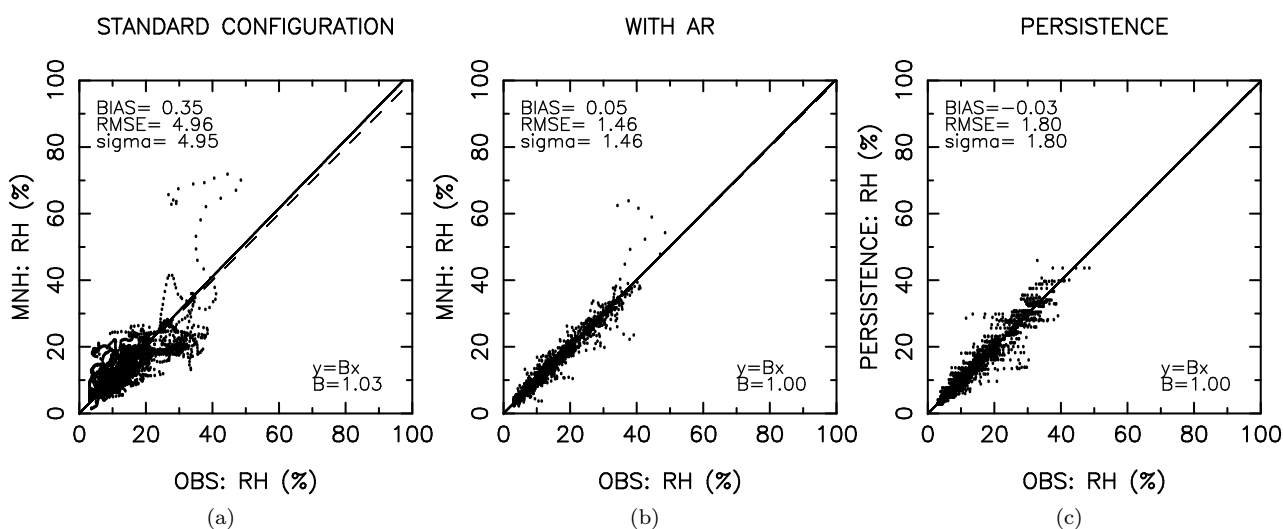


Figure 10: day1 - relative humidity (percent): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

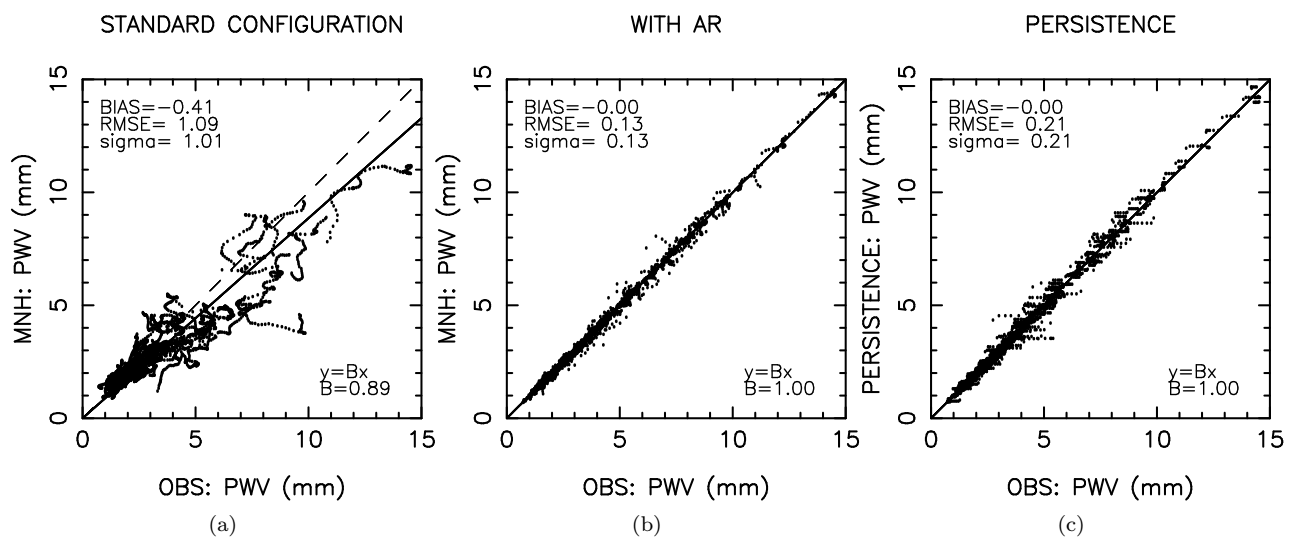


Figure 11: day1 - precipitable water vapor (mm): (a) STANDARD CONFIGURATION, (b) WITH AR (1H), (c) PERSISTENCE (1H)

2.2 Title for Statistics of Day 1

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VARIABLE	BIAS (inc. month)	RMSE (inc. month)	SD (inc. month)	BIAS (last month)	RMSE (last month)	SD (last month)
WS ($\text{m}\cdot\text{s}^{-1}$)	-0.22	2.48	2.47	-0.69	2.59	2.50
WD (degree)	-12.13	21.78	18.09	-12.93	20.91	16.44
RH (percent)	0.35	4.96	4.95	0.07	5.49	5.49
PWV (mm)	-0.41	1.09	1.01	-0.62	1.37	1.23
SEE (arcsec)	-	-	-	-	-	-
TAU (ms)	-	-	-	-	-	-
GLF (dimensionless)	-	-	-	-	-	-

Table 16: Statistics for variables in standard configuration: incremental month (i.e., since the beginning of service) *vs* last month (October-2023)

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VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.05	0.81	0.81		1.00	0.99
WD (degree)	0.15	5.48	5.47		9.55	9.11
RH (percent)	0.06	1.28	1.28		2.00	2.00
PWV (mm)	-0.00	0.15	0.15		0.23	0.23
SEE (arcsec)	-	-	-		-	-
TAU (ms)	-	-	-		-	-
GLF (dimensionless)	-	-	-		-	-

Table 17: Last month (October-2023): statistics for variables with AR (1H) *vs* persistence (1H)

VARIABLE	BIAS AR (1H)	RMSE AR (1H)	SD AR (1H)		RMSE persistence (1H)	SD persistence (1H)
WS ($m \cdot s^{-1}$)	-0.02	0.78	0.78		1.01	1.00
WD (degree)	0.01	6.34	6.34		11.45	11.07
RH (percent)	0.05	1.46	1.46		1.80	1.80
PWV (mm)	-0.00	0.13	0.13		0.21	0.21
SEE (arcsec)	-	-	-		-	-
TAU (ms)	-	-	-		-	-
GLF (dimensionless)	-	-	-		-	-

Table 18: Incremental month (i.e., since the beginning of service): statistics for variables with AR (1H) *vs* persistence (1H)

PARAMETER	STANDARD	WITH AR (1H)
POD1	32.5	87.3
POD2	76.4	88.1
POD3	82.5	95.6
PC	72.1	91.8
EBD	0.9	0.0

Table 19: day1 - PODs for wind speed (ms^{-1})

PARAMETER	STANDARD	WITH AR (1H)
POD1	96.4	98.4
POD2	54.5	88.3
POD3	30.1	87.0
PC	84.7	95.8
EBD	0.6	0.0

Table 20: day1 - PODs for wind speed (ms^{-1}) and thresholds (12.0,18.0) ms^{-1}

2.3 Title for PODs tables of Day 1

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PARAMETER	STANDARD	WITH AR (1H)
POD1	76.1	96.5
POD2	44.5	85.5
POD3	93.4	97.4
POD4	80.7	91.6
PC	76.8	95.1
EBD	0.0	0.0

Table 21: day1 - PODs for wind direction (degree)

PARAMETER	STANDARD	WITH AR (1H)
POD1	47.3	89.6
POD2	76.7	94.2
POD3	81.8	95.9
PC	72.9	93.8
EBD	1.1	0.0

Table 22: day1 - PODs for relative humidity (percent)

PARAMETER	STANDARD	WITH AR (1H)
POD1	99.6	99.8
POD2	NaN	NaN
POD3	NaN	NaN
PC	99.6	99.8
EBD	0.1	0.0

Table 23: day1 - PODs for relative humidity (percent) and thresholds (50,70) percent

PARAMETER	STANDARD	WITH AR (1H)
POD1	80.2	96.4
POD2	80.3	97.8
POD3	73.7	97.7
PC	77.9	97.4
EBD	0.1	0.0

Table 24: day1 - PODs for precipitable water vapor (mm)

3 WP_ESO_obs

Table 25 reports the status of success rate in carrying out a complete forecast. Different potential sources of failures related to the operations are taken into account. Tables 25 monitors the robustness of the automatic forecast system.

Each item in the first column starts with a label that indicates the origin of the 'potential' problem. Labels are: ECMWF, ESO, FATE. The success rate for the current month is calculated by the number of success cases divided by the number of days in a month. The success rate for the current incremental month is calculated by the number of success cases divided by the number of days in the number of months covered since the operation starting date up to the current month. The operations starting date is YYYYMMDD. The success rate is calculated using Eq.1:

$$1 - M/E \quad (1)$$

where M is the number of failures and E is the number of full days.

LEGENDA

- **ECMWF - ECMWF initialisation data transmission:** missing initialisation data coming from ECMWF that are supposed to be present in the servers 1a¹ and 1b in specific temporal windows on 24h. Server 1a is the main server located in ESO-Santiago (Chile), server 1b is the back-up server located in ESO-Garching (Germany).

- **ESO: access ESO - INPUT:** problem in the network connection. We consider a failure = 1 when initialisation data can not be downloaded in the temporal windows dedicated for carrying out the forecast for a full day (24h time scale) (contract Art.7.1).

- **ESO: access ESO - OUTPUT:** problem in the network connection. We consider a failure = 1 when delivery can not be upload on the main and back-up server within the time indicated in the technical specification. More precisely: two hours before the sunrise and the sunset (contract Art.7.1).

- **FATE: networks:** failure on networks (main and back-up) in LaMMA. We consider a failure = 1 when the main network and the back-up network have problems that do not allow the expected delivery on a full day.

- **ESO server INPUT problem:** server presenting bad operation. We have a failure = 1 when the server receiving the ECMWF data-set presents problems on a time scale of a full day.

- **ESO - server OUTPUT problem:** server presenting bad operation. We have a failure = 1 when the server receiving the delivery data-set from FATE presents problems on a time scale of a full day.

- **FATE - delivery failure in providing deliverables:** missing deliveries. We have a failure = 1 when deliveries are not uploaded for a full day (contract Art. 7.1).

Tables XX -Table XX show the percentage of real-time observations retrieved from ESO archive daily in real-time calculated on the last month and on the whole period of the operational service starting from the starting date of operations up to the current month. The estimation is done for the night time (Table XX) and day time (Table XX).

The percentage is obtained through Eq.X1 where M are the missing observations and E the expected observations. The expected observations are calculated considering the measurements ideally extended between

¹See tech-note-20221019-001

FATE - Monthly Report

the sunset and sunrise (night) and sunrise and sunset (day) with resampled data on 5 minutes time scale. The choice of 5 minutes is the largest possible $\Delta(T)$ that allows to provide a forecast at short time scale (AR) as conceived in this project.

FATE - Monthly Report

Automatic forecast system	Current month (success rate in %)	Incremental months (success rate in %)
ECMW: ECMWF initialisation data transmission	100.0	100.0
ESO: ftp access ESO - INPUT	100.0	100.0
ESO: ftp access ESO - OUPUT	100.0	100.0
FATE: principal network	100.0	100.0
ESO: server INPUT	100.0	100.0
ESO: server OUTPUT	100.0	100.0
FATE: hydrodynamic computation efficiency	100.0	100.0
FATE: delay on providing a forecast	100.0	100.0

Table 25: FATE: automatic forecasting system performance - statistics

TODO	Current month (success rate %)	Incremental months (success rate %)
SEE	89.200	86.400
TAU	88.400	83.500
GLF	90.000	86.600
WS	100.000	100.000
WD	100.000	100.000
RH	99.800	96.900
PWV	49.900	48.100

Table 26: night1 - real-time observations availability

TODO	Current month (success rate %)	Incremental months (success rate %)
WS	100.000	100.000
WD	100.000	100.000
RH	100.000	99.900
PWV	98.800	97.200

Table 27: day1 - real-time observations availability